

Pin Numbering - Raspberry Pi 3B+

- [Numbering Scheme](#)
- [Expansion Header](#)
- [J8 Pinout \(40-pin Header\)](#)
- [Known Issues](#)
- [Additional Resources](#)

Numbering Scheme

Pi4J (by default) uses an abstract pin numbering scheme to help insulate software from hardware changes. Pi4J implements the same pin number scheme as the WiringPi project. More information about the WiringPi pin number scheme can be found here: <http://wiringpi.com/pins/>

Pi4J provides a [RaspiPin](#) enumeration that is used to manage the accessible GPIO pins.

(NOTE: Pi4J also can be configured to use the Broadcom Pin numbering scheme.)

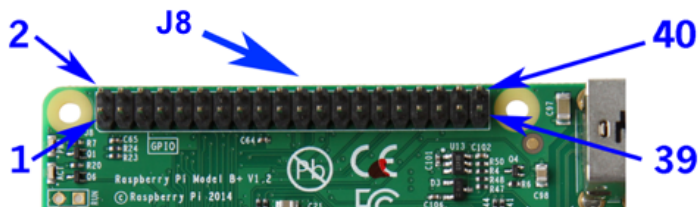
Please see this page for more information on both the WiringPi and Broadcom pin numbering schemes:

>> [Pin Numbering Schemes](#)

Expansion Header

The Raspberry Pi 3B+ board contains a single 40-pin expansion header labeled as 'J8' providing access to 28 GPIO pins.

(Pins 1, 2, 39 & 40 are also labeled below.)



J8 Pinout (40-pin Header)

The diagram below illustrates the GPIO pinout using the Pi4J/WiringPi GPIO numbering scheme.

Raspberry Pi 3 Model B (J8 Header)					
GPIO#	NAME			NAME	GPIO#
	3.3 VDC Power	1		2	5.0 VDC Power
8	GPIO 8 SDA1 (I2C)	3		4	5.0 VDC Power
9	GPIO 9 SCL1 (I2C)	5		6	Ground
7	GPIO 7 GPCLK0	7		8	GPIO 15 TxD (UART)
	Ground	9		10	GPIO 16 RxD (UART)
0	GPIO 0	11		12	GPIO 1 PCM_CLK/PWM0
2	GPIO 2	13		14	Ground
3	GPIO 3	15		16	GPIO 4
	3.3 VDC Power	17		18	GPIO 5
12	GPIO 12 MOSI (SPI)	19		20	Ground
13	GPIO 13 MISO (SPI)	21		22	GPIO 6
14	GPIO 14 SCLK (SPI)	23		24	GPIO 10 CE0 (SPI)
	Ground	25		26	GPIO 11 CE1 (SPI)
30	SDA0 (I2C ID EEPROM)	27		28	SCL0 (I2C ID EEPROM)
21	GPIO 21 GPCLK1	29		30	Ground
22	GPIO 22 GPCLK2	31		32	GPIO 26 PWM0
23	GPIO 23 PWM1	33		34	Ground
24	GPIO 24 PCM_FS/PWM1	35		36	GPIO 27
25	GPIO 25	37		38	GPIO 28 PCM_DIN
	Ground	39		40	GPIO 29 PCM_DOUT

Attention! The GPIO pin numbering used in this diagram is intended for use with WiringPi / Pi4J. This pin numbering is not the raw Broadcom GPIO pin numbers.

<http://www.pi4j.com>

Attention! The GPIO pin numbering used in this diagram is intended for use with WiringPi / Pi4J. This pin numbering is not the raw Broadcom GPIO pin numbers.

<http://www.pi4j.com>

([click here for hi-resolution image](#))

Known Issues

On the Raspberry Pi Model 3B+ the hardware-based serial/UART device `/dev/ttyAMA0` has been re-purposed to communicate with the built-in Bluetooth modem and is no longer mapped to the serial RX/TX pins on the GPIO header. Instead, a new serial port `/dev/ttyS0` has been provided which is implemented with a software-based UART (miniUART). This software-based UART (`/dev/ttyS0`) does not support PARITY and some have experienced some stability issues using this port at higher speeds. If you don't need Bluetooth functionality, you can disable the BT modem and configure the RPi to use a device-tree overlay to re-map the hardware-based serial UART (`/dev/ttyAMA0`) back to the GPIO header pins for TX/RX. See the instructions on this page for details on how to configure the device-tree overlay and disable the bluetooth modem/service:

[Disable Bluetooth Modem](#)

Additional Resources

- Please visit the [usage](#) page for additional details on how to control these pins using Pi4J.
- [Click here for more information on the Raspberry Pi Model 3B+.](#)
- [Click here for more information the Raspberry Pi pin functions.](#)